

MoE-LC: General-Purpose Lossless Compression for Multi-modal Data via Entropy-Aware Multi-Experts

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Abstract

The web-scale surge of multimodal content, including short-video feeds and autonomous sensing streams, has made web-native lossless compression a prerequisite for delivery and storage across browsers and edge-cloud pipelines. However, existing methods often fail to adapt to shifting distributions across different batches and struggle to balance computational resources in the face of large conditional entropy disparities among diverse modalities. To address these limitations, we propose *MoE-LC*, a new mixture-of-experts framework for multi-modal lossless compression that dynamically accommodates heterogeneous data distributions and varying complexity levels. First, the *Batch-Adaptive Experts* (BAE) module introduces batch-specific parameters with a residual gating mechanism, ensuring stable modeling under non-stationary distributions. Second, the *Entropy-Aware Multi-Expert Selection* (MES) strategy adaptively allocates the number of experts according to the data's estimated compression difficulty (entropy), thereby improving resource utilization and computational efficiency. Finally, the *Precision-Aware Expert Routing* (PER) component applies high-precision computation solely to the most critical experts, significantly reducing overhead without sacrificing compression accuracy. Experimental results across multiple real-world datasets demonstrate that **MoE-LC** achieves **5.33%–70.89%** improvements in compression ratio and **37.25%–1532.41%** gains in throughput compared to advanced baselines, offering a scalable solution for real-time, large-scale multi-modal data compression. Our code is available at https://github.com/Magie0/MoE_LC.

CCS Concepts

• Information systems → Information storage systems.

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Keywords

Lossless compression, multi-modal data, mixture of experts, conditional entropy, adaptive modeling, batch heterogeneity

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1 Introduction

The rapid surge of multi-modal applications—spanning short video streaming, immersive virtual reality, and autonomous driving—has led to an unprecedented expansion in data volume [4–7, 39]. In such large-scale, high-velocity settings, *lossless* compression is pivotal for both efficient data storage and transmission, as it preserves the exact original content during decompression. While classical frequency-based schemes constitute the backbone of many lossless approaches [11, 12, 15, 37], they often fail to keep pace with the complexity and diversity inherent in emerging multi-modal datasets. Meanwhile, contemporary deep-learning-based models [31, 33–35] can offer significant compression benefits but frequently struggle under heterogeneous, rapidly changing data distributions across different modalities or time steps.

Figure 1 highlights the two principal challenges hampering progress in large-scale multi-modal lossless compression:

- *Cross-batch distribution heterogeneity*: Even within a single modality, data statistics can shift sharply from batch to batch, causing fixed-parameter models to adapt poorly and degrade in performance.
- *Conditional-entropy disparities*: When combining text, images, and other domains, the compressibility of each modality varies considerably. Data with lower entropy (e.g., repetitive text) can benefit greatly from additional modeling capacity, whereas high-entropy data (e.g., high-resolution images) may yield minimal improvements despite larger ranges.

To address these concerns, we propose **MoE-LC** (*Mixture-of-Experts Lossless Compression*), an **online** framework for handling batch-wise distribution shifts and heterogeneous data modalities. MoE-LC is built around three core components that operate in

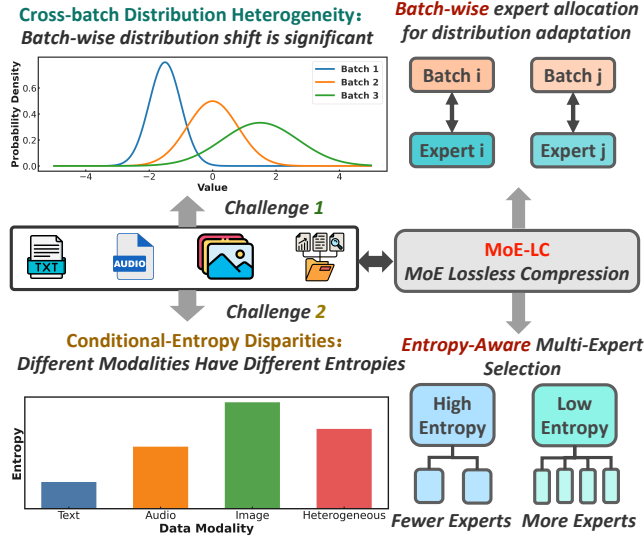


Figure 1: Overview of the core challenges addressed by MoE-LC. (1) Cross-batch distribution heterogeneity: rapid changes in batch-wise data distributions require adaptive parameterization. (2) Conditional-entropy disparities: varying entropy levels across different modalities demand dynamic expert allocation. Our framework tackles these issues through batch-specific parameter generation, entropy-aware expert selection, and precision-controlled routing.

synergy. First, *Batch-Adaptive Experts (BAE)* provides a per-batch parameterization mechanism, generating specialized parameters for each incoming batch while using a residual gating strategy to ensure rapid adaptation and stability. Second, *Entropy-Aware Multi-Expert Selection (MES)* employs a difficulty metric based on cross-entropy loss and conditional entropy to determine how many experts should be active for each input, allocating additional modeling capacity to lower-entropy data while conserving resources for higher-entropy data. Finally, *Precision-Aware Expert Routing (PER)* refines the computational efficiency by ranking the selected experts and deploying high-precision computation (e.g., FP32) only for those deemed most relevant, while assigning lower-precision (e.g., FP16) resources to experts of lesser impact. Together, these components enable MoE-LC to adaptively balance modeling power, numerical precision, and computational cost across diverse and evolving data distributions.

By jointly tackling cross-batch heterogeneity and modality-specific compressibility, MoE-LC achieves both adaptability and efficiency. Empirical evaluations reveal that our framework consistently outperforms advanced baselines in terms of compression ratio and throughput, achieving **5.33%–70.89%** better compression and **37.25%–1532.41%** higher throughput. These results underscore the importance of allocating modeling resources in a context-dependent manner. In summary, the main contributions of this work are:

- We propose a new online lossless compression framework, *MoE-LC*, that dynamically addresses batch-level distribution

shifts and varying conditional entropy across different data domains.

- We develop a three-tiered pipeline consisting of BAE, MES, and PER, enabling flexible control over modeling capacity and numerical precision for more efficient and accurate compression.
- We validate *MoE-LC* on a range of real-world datasets, demonstrating clear gains in both compression ratio and system throughput compared to state-of-the-art baselines.

2 Related Work

2.1 Entropy in Lossless Compression

Entropy, as defined by Shannon [42, 43], measures the average uncertainty in a data source and provides the theoretical lower bound for lossless compression [1, 8, 48]. For a discrete random variable X with probability mass function $p(x)$, the entropy is defined as:

$$H(X) = - \sum_{x \in \mathcal{X}} p(x) \log p(x). \quad (1)$$

Traditional algorithms such as Huffman coding [20] and arithmetic coding [50] exploit this principle to construct efficient codes. Recent neural compression models [13, 45] incorporate learned entropy models to guide bit allocation more precisely.

In multi-modal settings, entropy also reflects data complexity: low-entropy data (e.g., text) tends to be more compressible, while high-entropy data (e.g., images or sensor signals) is more difficult to model. This motivates entropy-aware mechanisms that adapt compression strategies based on data characteristics—an idea we operationalize in our expert selection module.

2.2 Mixture-of-Experts Architectures

The Mixture-of-Experts (MoE) architecture [22] is a classical divide-and-conquer strategy that partitions the input space and assigns different sub-models, or “experts,” to process different types of data patterns. A gating network dynamically routes each input to one or a subset of experts, allowing the model to maintain high expressiveness with sparse activation. This sparsity is particularly appealing for scaling model capacity without a proportional increase in computational cost. In recent years, MoE has gained renewed interest in the deep learning community, especially in large-scale language modeling. Notably, works such as GShard [28] and Switch Transformer [16] leverage sparse expert routing to scale Transformer models to trillions of parameters while keeping training and inference costs manageable. These advances highlight the effectiveness of MoE in dynamically allocating model capacity according to the characteristics of the input.

In the context of compression, however, MoE remains underexplored. Some recent studies [29] have begun to investigate expert-based neural compression, typically in the context of image or video coding. These works show that different experts can specialize in capturing different structural or semantic patterns, thereby improving compression quality. Nonetheless, most existing designs assume static routing or fixed expert numbers, which limits adaptability in dynamic, heterogeneous scenarios such as multi-modal lossless compression.

2.3 Deep Learning for Lossless Compression

Recent advances in deep learning have given rise to a new generation of lossless compression methods, which can be broadly categorized based on whether model parameters are updated during compression: **static**, **dynamic**, and **semi-dynamic**.

Static methods train predictive models *offline* and fix their parameters during inference. Representative approaches such as DeepZip [17] and LLMZip [45] leverage RNNs or large language models to estimate token probabilities. While these models can offer strong performance on in-distribution data, they often struggle to generalize to unseen inputs and require expensive pre-training. Moreover, using large static models across diverse domains typically leads to high inference latency and increased memory consumption.

Dynamic methods, including CMIX [25] and NNCP [2], initialize models from scratch and continually update parameters *online* during compression. This enables them to adapt to previously unseen distributions and data structures without requiring prior training. SEP[47] enhances compression performance through semantic dependency modeling while accelerating throughput via multi-stream parallel processing. However, such methods may suffer from cold-start inefficiencies in early stages and tend to incur higher computational overhead due to frequent parameter updates [30, 31, 33–35].

Semi-dynamic approaches attempt to combine the strengths of both static and dynamic paradigms. For instance, DZip [18] begins with a pre-trained model and selectively updates auxiliary components during compression to improve adaptability. While this strategy reduces initialization cost and enhances flexibility, it still relies on a pre-trained backbone and additional parameter storage, which can become a bottleneck in resource-constrained environments.

Overall, despite notable progress, existing deep-learning-based lossless compression methods often fail to balance adaptability, efficiency, and compression quality—particularly in the presence of highly diverse or evolving data distributions. These limitations motivate the design of our *MoE-LC* framework, which achieves fine-grained adaptation and resource control without incurring the overhead of fully dynamic models or the rigidity of static ones.

3 Preliminary Observations

Real-world multimodal corpora exhibit two recurring phenomena that directly shape the design of general-purpose lossless compression systems: (i) *cross-batch distribution heterogeneity* within a single dataset and (ii) *entropy disparity across modalities*. These observations, summarized in Figure 2, explain why a practical system must balance code-length optimality against throughput under non-stationary and heterogeneous conditions.

Notation. Let $\mathcal{V}$ be the byte/token alphabet. For batch b with samples $X^{(b)} = \{x_i\}$, denote the empirical distribution by $p_b(v) = \frac{1}{|X^{(b)}|} \sum_{x_i \in X^{(b)}} \mathbf{1}[x_i = v]$ for $v \in \mathcal{V}$. Let q denote the predictive distribution used for coding. $H(\cdot)$ is Shannon entropy and $D_{\text{KL}}(\cdot \| \cdot)$ is Kullback–Leibler divergence.

3.1 Cross-Batch Distribution Heterogeneity

Empirical observation. Figure 2(a) shows that consecutive batches drawn from the same dataset (e.g., Silesia) often present markedly

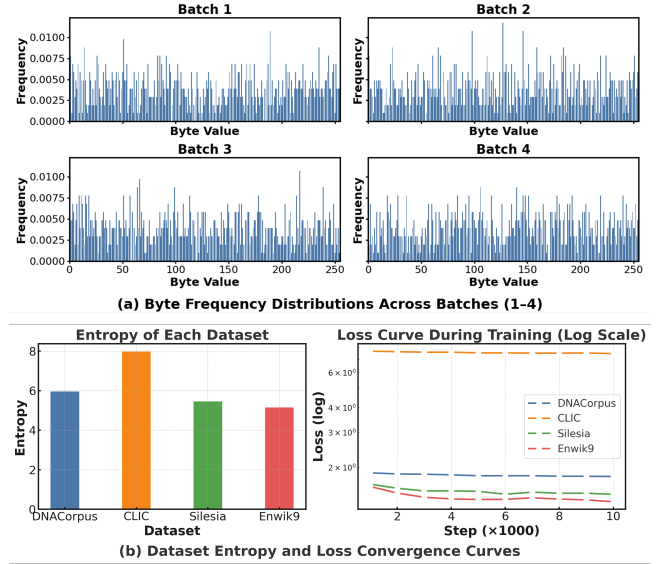


Figure 2: (a) Byte-level frequency distributions of the first four batches from the Silesia dataset, showing clear batch-wise distribution heterogeneity. (b) Entropy statistics and training loss curves across four datasets. The bar chart on the left illustrates entropy disparities among datasets, where CLIC exhibits the highest entropy and Enwik9 the lowest. The line plot on the right shows log-scaled loss values over training steps, demonstrating convergence behaviors across datasets.

different empirical profiles, indicating non-stationarity at the batch scale.

Implication for coding. When a static-parameter predictor q —fit on earlier data or an aggregate—encounters a new batch with distribution p_b , the expected code length per symbol obeys the classical mismatch identity:

$$\mathbb{E}_{x \sim p_b} [-\log q(x)] = H(p_b) + D_{\text{KL}}(p_b \| q).$$

Thus, even when $H(p_b)$ is moderate, a distributional mismatch (large D_{KL}) inflates code length, yielding *compression volatility* across batches in the absence of batch-level adaptation.

3.2 Entropy Disparity Across Modalities

Empirical observation. Figure 2(b) compares representative datasets and reveals a pronounced spread in entropy: image corpora (e.g., CLIC) occupy the high-entropy end, while large-scale text (e.g., Enwik9) is substantially lower. Correspondingly, training loss curves typically converge faster on low-entropy data and slower—often plateauing—on high-entropy data, even with larger-capacity models.

Capacity allocation policy. To jointly optimize compression ratio and throughput under a fixed compute budget, capacity should be *allocated non-uniformly across modalities*:

- *Low-entropy modalities: allocate more experts/capacity.* They exhibit stable, reusable structure (syntax, templates, frequent

patterns). Additional experts specialize on such structure and reduce cross-entropy rapidly, yielding large, reliable bit savings per unit compute.

- *High-entropy modalities: allocate fewer experts/capacity.* They approach noise-like or fine-grained variability. The marginal coding gain from extra experts diminishes quickly, while compute costs grow and throughput degrades. Conservatively limiting capacity prevents inefficient compute spending and reduces the risk of overfitting short-term idiosyncrasies.

This non-uniform policy targets *system-level* optimality: maximize sustainable bit savings per FLOP across modalities, rather than maximizing gains on any single, high-entropy source at the expense of overall throughput.

4 Method

To address these challenges, we introduce **MoE-LC**, a unified framework for online multimodal lossless compression. As illustrated in Figure 3, our method incorporates three complementary mechanisms: *Batch-Adaptive Experts (BAE)*, *Entropy-Aware Multi-Expert Selection (MES)*, and *Precision-Aware Expert Routing (PER)*—to adaptively match model capacity to data complexity at the batch, sample, and expert levels. Together, these components enable MoE-LC to achieve fine-grained adaptation, efficient resource allocation, and robust performance across diverse data modalities.

4.1 Batch-Adaptive Experts (BAE)

Online streams are batch-wise non-stationary; with static parameters, the predictor mismatches the current batch distribution and yields volatile code length[23]. **BAE** adds a small set of *batch-specific parameters* to the expert transformation and controls their effect with a residual gate, enabling rapid local adaptation with a safe fallback.

Per-batch parameterization. Let $\mathbf{X} \in \mathbb{R}^{B \times d}$ denote an input batch. For batch b , we attach a dedicated parameter set $\theta^{(b)}$ to the shared expert transformation and compute

$$\mathbf{A} = f_{\theta^{(b)}}(\mathbf{X}) \in \mathbb{R}^{B \times d} \quad (2)$$

Here $\theta^{(b)}$ is specific to batch b (i.e., not reused across batches). Its role is to let the expert adjust to batch-local statistics.

Residual gating mechanism. A scalar gate per sample moderates the contribution of the batch-specific expert:

$$\mathbf{g} = \sigma(\mathbf{X}\mathbf{W}_g + \mathbf{b}_g) \in \mathbb{R}^{B \times 1} \quad (3)$$

$$\mathbf{Y} = \mathbf{g} \odot \mathbf{A} + (1 - \mathbf{g}) \odot \mathbf{X} \in \mathbb{R}^{B \times d} \quad (4)$$

where $\mathbf{W}_g \in \mathbb{R}^{d \times 1}$ and $\mathbf{b}_g \in \mathbb{R}$ are *shared (global) gating parameters* learned with the model; $\sigma(\cdot)$ is the sigmoid and $\odot$ denotes broadcasted element-wise multiplication. When $g \approx 1$, the batch-specific path dominates; when $g \approx 0$, the model safely falls back to the base pathway $\mathbf{X}$. This residual design stabilizes training and prevents code-length regressions under distributional drift while preserving throughput[19].

4.2 Entropy-Aware Multi-Expert Selection (MES)

Uniformly activating the same number of experts across datasets or streams is either wasteful on easy, low-entropy sources or ineffective on hard, high-entropy ones[3, 44]. **MES** makes a *single, global* decision—once per dataset/stream—on how many experts to activate, so that capacity is aligned with intrinsic difficulty while preserving throughput.

Step 1: Full-model complexity probing. Before coding, we run the full-capacity predictor P_{full} with all E experts on a shared probe segment D_{probe} (visible to both encoder and decoder) to obtain per-symbol predictive distributions

$$\tilde{y}_{\text{full},i} = P_{\text{full}}(x_i | x_{<i}), \quad x_i \in D_{\text{probe}} \quad (5)$$

We then define the *global difficulty fingerprint* as the average predictive entropy (bits/symbol)

$$C_{\text{data}} = \mathbb{E}_{x_i \in D_{\text{probe}}} \left[- \sum_j \tilde{y}_{\text{full},i,j} \log_2 \tilde{y}_{\text{full},i,j} \right] \quad (6)$$

Using *the full model* for probing is crucial: it yields the most informative (least underfit) estimate of compressibility, avoids selection bias from any pre-pruned subset of experts, and gives a stable, reproducible signal that both sides can compute once and amortize over the whole stream.

Step 2: Sigmoid policy to choose the expert count. We map C_{data} to the number of active experts via a monotone policy

$$k^* = \left\lfloor E \cdot \sigma(-\beta C_{\text{data}} + \gamma) \right\rfloor, \quad 1 \leq k^* \leq E, \quad (7)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$, $\beta > 0$ controls sensitivity to entropy changes, and $\gamma \in \mathbb{R}$ sets the operating midpoint. Intuitively, lower entropy (richer reusable structure) justifies *more* experts because added capacity converts to large, stable bit savings per FLOP, whereas higher entropy (rapidly diminishing returns) calls for *fewer* experts to protect throughput and decoding cost. The parameters (β, γ) are fixed protocol hyperparameters rather than learned per stream.

Determinism and side-channel freedom. Since k^* is a deterministic function of D_{probe} , E , and (β, γ) , both encoder and decoder recover the same decision independently:

$$k^* = f_{\text{mes}}(D_{\text{probe}}; E, \beta, \gamma) \quad (8)$$

$$H(k^* | D_{\text{probe}}) = 0 \quad (9)$$

so no side information is transmitted. This single probe-and-decide step fixes the expert budget for the remainder of coding, eliminating per-token routing overhead while retaining dataset-level adaptivity.

4.3 Precision-Aware Expert Routing (PER)

Once MES determines the top- $k(x)$ experts for input x , we further streamline computation through **Precision-Aware Expert Routing (PER)**. Specifically, although MES selects the most relevant experts, it remains beneficial to allocate more computational resources (in the form of higher numerical precision) to a subset of those experts with the greatest impact on the final prediction, while assigning lower precision to others.

Input-Aware Expert Relevance Estimation. Let $\mathcal{E}(x)$ be the set of $k(x)$ experts activated by MES. To identify which of these experts plays a pivotal role for a particular input, we employ a

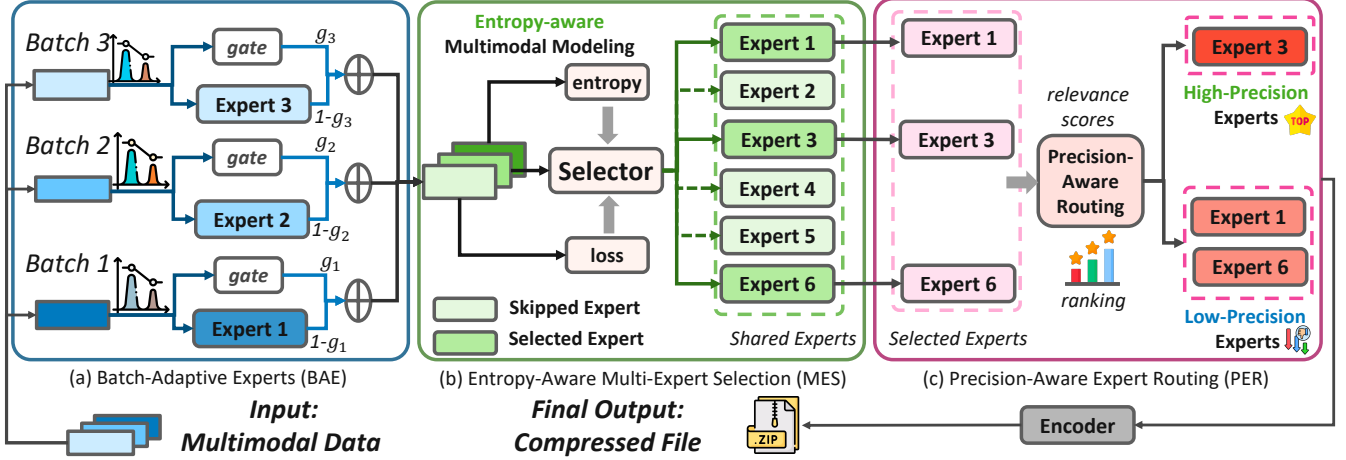


Figure 3: Overview of the proposed MoE-LC framework for multimodal lossless compression. The system comprises three key components: (a) *Batch-Adaptive Experts (BAE)* assigns a dedicated expert to each input batch and combines expert outputs with raw inputs via input-dependent residual gating, enabling robust adaptation to batch-wise distribution shifts. (b) *Entropy-Aware Multi-Expert Selection (MES)* computes the entropy and modeling loss of each input to determine a dynamic number of experts for fine-grained processing, selecting a subset of shared experts accordingly. (c) *Precision-Aware Expert Routing (PER)* ranks the selected experts by their relevance scores and assigns high or low numerical precision to each, minimizing computational overhead while preserving compression quality. The output from the precision-weighted expert ensemble is encoded into the final compressed file.

lightweight gating network that projects an input embedding $\mathbf{h} \in \mathbb{R}^{B \times d}$ onto a set of relevance scores:

$$\mathbf{s} = \text{softmax}(\mathbf{h} \mathbf{W}_{\text{gate}} + \mathbf{b}_{\text{gate}}) \in \mathbb{R}^{B \times k(x)}, \quad (10)$$

where $\mathbf{W}_{\text{gate}} \in \mathbb{R}^{d \times k(x)}$ and $\mathbf{b}_{\text{gate}} \in \mathbb{R}^{k(x)}$. This mechanism assigns an importance value to each expert in $\mathcal{E}(x)$ based on the input representation, enabling selective prioritization.

Precision-Level Routing. With these relevance scores $\mathbf{s}$ in hand, PER designates the top- r experts for computation in high precision (e.g., FP32):

$$\mathcal{E}_{\text{high}}(x) = \text{Top-}r(\{s_i(x)\}_{i \in \mathcal{E}(x)}), \quad (11)$$

while the remaining $\mathcal{E}_{\text{low}}(x)$ operate in lower precision (e.g., FP16):

$$\mathcal{E}_{\text{low}}(x) = \mathcal{E}(x) \setminus \mathcal{E}_{\text{high}}(x). \quad (12)$$

Higher-precision computation is reserved for experts that exhibit the strongest relevance scores, ensuring minimal performance degradation while reducing memory overhead and computational cost for the less critical experts. This fine-grained control over numerical precision further enhances the efficiency of the overall mixture-of-experts framework, enabling robust yet resource-conscious compression across diverse data regimes.

4.4 Overall Compression Pipeline

Algorithm 1 provides a concise overview of our online multimodal lossless compression process under the MoE-LC framework. First, each batch of input data undergoes *per-batch parameterization* through the Batch-Adaptive Experts (BAE), allowing the model to capture unique distributional nuances that may arise from rapidly evolving data streams. Next, the Entropy-Aware Multi-Expert Selection (MES) module decides how many experts should be assigned

Algorithm 1: MoE-LC Procedure for Adaptive Lossless Compression

Input: Input byte stream $X = \{x_1, x_2, \dots, x_T\}$
Output: Compressed bitstream

- 1 Split X into batches $\{X^{(1)}, X^{(2)}, \dots, X^{(B)}\}$;
- 2 **foreach** batch $X^{(b)}$ **do**
 - // Batch-wise parameterization (BAE)
 - 3 Generate expert weights $\theta^{(b)}$ for $X^{(b)}$;
 - // Expert selection (MES)
 - 4 Estimate difficulty $\Phi(X^{(1:B)})$ and determine k experts;
 - // Precision routing (PER)
 - 5 Rank expert relevance and assign precision accordingly;
 - // Prediction and compression
 - 6 Compute $P(x_i | x_{1:n:i-1})$ for all $x_i \in X$;
 - 7 Encode each x_i using arithmetic coding with $P(x_i | x_{1:n:i-1})$;
- 8 **return** Compressed bitstream

to each sample by assessing its estimated difficulty. This not only streamlines computational usage but also preserves high modeling capacity for inherently complex or high-entropy inputs. Subsequently, Precision-Aware Expert Routing (PER) refines the selected experts further by distributing numerical precisions (e.g., FP32 or FP16) according to expert-level relevance scores, thereby minimizing resource usage without sacrificing predictive accuracy.

Once the final set of experts and corresponding precision levels is established, the model outputs a predicted distribution for each

input token or byte, which is then compressed via arithmetic coding to generate the final bitstream. Crucially, the entire pipeline operates in an *online* fashion, adapting as new batches arrive and ensuring that no single module becomes a bottleneck for overall system throughput.

5 Experiment

5.1 Experimental Setup

Environment. All experiments were conducted on a workstation equipped with 256 GB of memory and a single NVIDIA RTX 3080Ti GPU with 12 GB of VRAM. To ensure a fair comparison, all baselines and our model were executed using only one GPU.

Datasets. We evaluate MoE-LC across 12 well-established benchmark datasets, which span a wide range of modalities including text, image, audio, genome, and mixed-type corpora. Detailed descriptions and dataset statistics are provided in Table 1.

Baselines. Our method is compared with six traditional lossless compressors—such as Gzip [15], PBZIP2 [38], Snzip [27], LZ4 [11], LZMA2 [37], and PPMD [9]—and six deep learning-based approaches including DeepZip [17], lstm-compress [26], DZip [18], TRACE [34], PAC [35] and MSDZip [31]. All baseline results are obtained using their official codebases and default parameter configurations as described in the corresponding papers.

Metrics. To assess both compression effectiveness and efficiency, we report two standard metrics: *Compression Ratio (CR)* and *Throughput (THP)*.

The compression ratio measures the number of bits needed to represent each byte of original input:

$$CR = \frac{Size_{compressed}}{Size_{source}} \times 8 \quad (\text{bits/base}) \quad (13)$$

where $Size_{compressed}$ and $Size_{source}$ represent the compressed file size and the original input size, respectively. A smaller CR value indicates better compression performance.

The throughput reflects the overall speed of the compression-decompression process:

$$THP = \frac{Size_{source}}{CT + DT} \quad (\text{bytes/second}) \quad (14)$$

where CT and DT denote the time spent on compression and decompression, respectively. Higher THP values are preferred.

MoE-LC Settings. The shared expert network in MoE-LC uses a two-layer feedforward structure, with hidden sizes set to 512 and 4096, respectively. We fix the number of modeled input bytes to 16 across all experiments. Unless otherwise specified, our configuration follows the standard settings used by prior neural compression models. To evaluate scalability, we vary the batch size from 512 to 8192. All experiments for MoE-LC are conducted using a single GPU [24].

5.2 Compression Ratio

Table 2 reports the compression ratios (bits/base) of our method MoE-LC and 12 strong baselines, including both traditional and deep learning-based compressors, across 12 datasets covering diverse modalities such as text, image, audio, genome, and heterogeneous data. Compared to all existing compressors, MoE-LC consistently achieves the lowest compression ratio across all datasets.

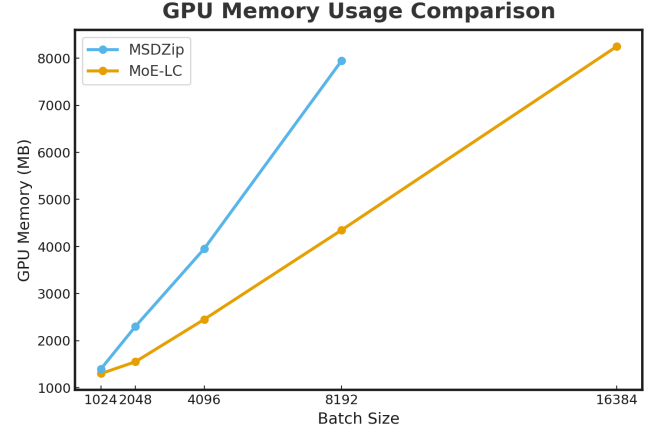


Figure 4: GPU memory usage comparison between MSDZip and our MoE-LC across different batch sizes. MoE-LC achieves significantly lower memory consumption under large batch settings, demonstrating better scalability.

Notably, MoE-LC yields the best performance on high-entropy datasets like CLIC (image) and LJSpeech (audio), demonstrating its robustness in complex scenarios. On low-entropy textual datasets such as Enwik9 and Text8, MoE-LC still maintains superior efficiency with minimal bits per base.

Overall, MoE-LC achieves an average compression ratio of **2.13 bits/base**, outperforming the previous state-of-the-art MSDZip (2.25) by 5.33%. In addition, MoE-LC reduces the compression ratio compared to PAC by 6.66%, TRACE by 12.24%, and DZip by 9.98%. Traditional compressors show even larger gaps: MoE-LC improves over Gzip by 39.73%, over PBZIP2 by 30.12%, and over Snzip by a remarkable 55.63%. These results validate the effectiveness of our entropy-aware expert selection and precision routing mechanisms, and clearly demonstrate that MoE-LC not only generalizes well across modalities but also sets a new benchmark for multi-modal lossless compression.

5.3 Compression Throughput

We assess the efficiency of MoE-LC from two perspectives: compression throughput and GPU memory usage.

Table 3 summarizes the compression throughput (bytes/second) of our proposed MoE-LC compared to six representative baseline methods across twelve datasets of diverse modalities. Experimental results show that **MoE-LC achieves the highest average throughput among all methods**, significantly surpassing existing learning-based compressors such as MSDZip and DZip, with 2.34× and 9.54× improvements, respectively.

Compared to other approaches, MoE-LC demonstrates strong adaptability across different modalities—including text, image, audio, and genome data—by dynamically adjusting the number of experts and precision strategies according to data complexity. Benefiting from its sparse activation and precision-aware computation mechanisms, MoE-LC greatly reduces computational redundancy while maintaining competitive compression performance, validating its generality and efficiency in multi-modal compression scenarios.

Table 1: Detailed information of all datasets

Index	Dataset	Type	Size (bytes)	Description
D1	Enwik8 [32]	text	100000000	First 10^8 bytes of the English Wikipedia dump on 2006
D2	Text8 [32]	text	100000000	First 10^8 bytes of the English Wikipedia (only text) dump on 2006
D3	Enwik9 [32]	text	1000000000	First 10^9 bytes of the English Wikipedia dump on 2006
D4	Book [46]	text	1000000000	First 10^9 bytes of BookCorpus
D5	Silesia [14]	heterogeneous	211938580	A heterogeneous corpus of 12 documents with various data types
D6	Backup [34]	heterogeneous	1000000000	10^9 bytes random extract from the disk backup of TRACE
D7	CLIC [10]	image	243158876	Classical image compression benchmark (validation) of the CLIC 2024
D8	ImageTest [41]	image	470611702	A new 8-bit benchmark dataset for image compression evaluation
D9	GoogleSpeech [49]	audio	327759206	First 10,000 audio files of the Google Speech Commands Dataset
D10	LJSpeech [21]	audio	293847664	First 10,000 audio files of the LJSpeech Dataset
D11	DNACorpus [40]	genome	685597124	A corpus of DNA sequences from 15 different species
D12	GenoSeq [36]	genome	1926041160	A collection of genomics sequencing dataset with FastQ format

Table 2: Compression Ratio (bits/base) of our MoE-LC and baselines. The Boldface indicates the best result.

Compressor	text				heterogeneous		image		audio		genome		Overall	Imp. (%)
	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12		
Traditional MSD Compressor														
Gzip	2.916	2.644	2.581	2.875	2.554	6.237	7.992	5.885	5.770	6.851	2.171	1.616	3.534	39.728
PBZIP2	2.321	2.112	2.033	2.143	2.063	5.978	7.977	4.643	4.754	5.872	2.103	1.324	3.048	30.118
Snzip	4.475	4.303	4.020	4.678	3.829	7.100	8.001	7.326	7.387	7.983	3.682	2.676	4.800	55.625
LZMA2	1.989	1.858	1.718	2.027	1.839	5.084	7.956	4.786	4.717	5.850	1.800	1.198	2.799	23.901
PPMD	1.853	1.712	1.602	1.801	1.816	5.706	7.956	4.437	4.453	5.587	1.944	1.209	2.805	24.064
LZ4	3.359	3.117	2.980	3.263	2.921	6.653	8.000	6.658	6.514	7.723	2.734	1.977	3.985	46.550
Learning-based MSD Compressor														
lstm-compress	1.854	1.756	1.571	1.704	1.776	4.475	7.505	7.225	3.623	3.623	7.605	0.934	3.156	32.510
DeepZip	1.952	1.806	18.027	1.681	1.870	4.340	22.204	20.272	14.398	14.398	1.863	1.018	7.316	70.886
DZip	1.875	1.757	1.539	1.591	1.716	4.236	8.021	3.595	3.805	3.805	1.799	0.908	2.366	9.975
TRACE	1.870	1.782	1.556	1.782	1.771	4.547	7.757	3.496	3.664	3.664	1.870	0.988	2.427	12.237
PAC	1.695	1.626	1.377	1.590	1.604	4.200	7.507	3.362	3.869	3.869	1.802	0.849	2.282	6.661
MSDZip	1.692	1.624	1.375	1.583	1.593	4.174	7.579	3.345	3.411	4.534	1.784	0.847	2.25	5.333
MoE-LC (Ours)	1.588	1.546	1.244	1.512	1.507	4.083	7.578	3.339	3.374	4.293	1.773	0.636	2.13	–

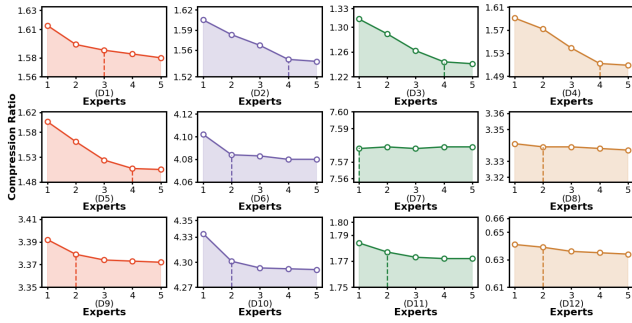
Figure 4 illustrates the GPU memory usage of our proposed MoE-LC compared to the prior state-of-the-art MSDZip under varying batch sizes. As the batch size increases, the memory requirement of MSDZip grows steeply, reaching over 12 GB for a batch size of 16384. In contrast, MoE-LC shows a slower growth rate and remains within 8 GB up to a batch size of 8192. This improvement stems from our entropy-aware expert selection and precision routing, which reduce unnecessary parameter activation and computation. These results validate the superior memory efficiency and scalability of MoE-LC, making it more suitable for large-batch or real-time compression scenarios.

5.4 Ablation Study

Table 4 presents the ablation study of our proposed MoE-LC framework on the Enwik8 dataset. We evaluate the effect of each core module—Batch-Adaptive Experts (BAE), Entropy-Aware Multi-Expert Selection (MES), and Precision-Aware Expert Routing (PER)—by removing one component at a time. Removing BAE (row 2) leads to a significant drop in compression performance, increasing the compression ratio from **1.588** to **1.768**, indicating that static parameterization struggles to capture distribution shifts across batches. Eliminating MES (row 3) also causes a performance degradation, raising the compression ratio to **1.733**, which highlights the necessity of adaptive expert allocation for balancing efficiency and accuracy.

Table 3: Throughput (bytes/second) of MoE-LC and baselines across all datasets. The Boldface means the best result. MoE-LC is highlighted in light blue.

Compressor	text				heterogeneous		image		audio		genome		Overall	Imp. (%)
	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12		
lstm-compress	4964	7137	4616	4586	4198	4393	4304	3968	4492	3969	8028	7137	5149	1532.41
DeepZip	14398	10753	14933	14777	14338	12421	13998	12487	9213	13333	20951	18553	14180	492.76
DZip	8619	7727	10770	7902	8371	8047	9362	10052	5280	8901	9053	11628	8809	854.17
TRACE	30630	31304	30105	30749	30909	29791	30940	30045	29829	31639	31674	31352	30747	173.37
PAC	61040	60616	61817	61317	60666	61430	60486	62278	60946	61773	61054	61481	61242	37.25
MSDZip	35936	35195	36802	36403	36740	36649	36040	36040	34997	35217	34909	35482	35923	133.98
MoE-LC	76800	56320	56320	56320	56320	97280	122880	97280	97280	97280	97280	97280	84053	–

**Figure 5: Compression ratio curves with respect to the number of experts on 12 datasets (D1–D12). The dashed line indicates the expert number selected by our MES. Compression ratio tends to stabilize after the selected point.****Table 4: Ablation study of the MoE-LC framework on Enwik8. We remove one component at a time to evaluate its impact on compression performance.**

ID	Variant	CR ↓	THP ↑	Mem ↓
(1)	MoE-LC (Full)	1.588	76800	4372MB
(2)	w/o BAE	1.768	1365440	2736MB
(3)	w/o MES	1.733	1227900	3050MB
(4)	w/o PER	1.580	59800	4924MB

Disabling PER (row 4) only slightly affects the compression ratio, but increases memory usage from **4372MB** to **4924MB**, reflecting its role in reducing computational overhead through selective precision control. Overall, the full MoE-LC framework achieves the best trade-off between compression ratio, throughput, and memory, confirming the complementary benefits of all three components.

5.4.1 Effectiveness of Entropy-Aware Multi-Expert Selection (MES). To evaluate the effectiveness of the proposed MES mechanism, we analyze its ability to dynamically select the optimal number of experts based on the estimated complexity of each input. Instead of using a fixed number of experts for all samples, MES leverages a combination of entropy and cross-entropy signals to allocate computation adaptively.

As shown in Figure 5, our model achieves the highest compression ratio when the number of active experts is adaptively determined by MES, rather than being statically fixed. This indicates that MES successfully identifies samples with lower conditional entropy—such as repetitive or structured data—and assigns them more expert capacity to fully exploit their compressibility. Conversely, it reduces the number of experts for high-entropy data, avoiding unnecessary computational cost.

6 Conclusion

In this work, we introduced **MoE-LC**, a new online framework for multi-modal lossless compression that adaptively handles both batch-wise distribution shifts and conditional-entropy disparities across modalities. To this end, MoE-LC incorporates three synergistic modules: *Batch-Adaptive Experts (BAE)* for per-batch parameterization by generating dynamic expert weights to adapt to shifting input distributions, *Entropy-Aware Multi-Expert Selection (MES)* for input-dependent expert allocation based on entropy-guided difficulty estimation, and *Precision-Aware Expert Routing (PER)* for computational efficiency via selectively assigning higher precision to more relevant experts while reducing overhead for others. Extensive experiments across 12 datasets covering diverse data types—text, image, audio, and genome—demonstrate that MoE-LC consistently outperforms traditional and deep-learning-based baselines in terms of compression ratio and throughput, while maintaining practical memory usage. These results highlight the effectiveness of modeling adaptation, entropy-aware resource scheduling, and precision control in enhancing both accuracy and efficiency for large-scale lossless compression.

We hope this work sheds light on the importance of dynamic modeling strategies in multi-modal settings and opens up future research directions in adaptive compression and efficient expert-based architectures.

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